PALANTIR C2

AI-Enabled Command & Control System

CAPABILITY BRIEF

Automated F2T2EA Kill Chain | Multi-Agent AI Orchestration
Coordinated Drone Swarm Operations | Real-Time Sensor Fusion

OPERATIONAL NEED

Current kill chain gaps put warfighters at risk

Sensor-to-Shooter Gap

Manual handoffs between ISR, targeting, and effects create 45-90 minute cycle times. Near-peer adversaries compress this to under 10 minutes.

Multi-Domain Integration

Sensor data from UAV, satellite, SIGINT, and ground assets remains siloed across incompatible systems. No unified common operating picture.

ISR Analyst Shortage

Demand for trained ISR analysts exceeds supply 4:1. AI augmentation is no longer optional — it is a force readiness imperative.

Accountability & Compliance

DoD Directive 3000.09 mandates human oversight for autonomous weapon systems. Systems must provide auditable decision trails.

SYSTEM OVERVIEW

Automated F2T2EA kill chain with AI orchestration

FIND	FIX	TRACK	TARGET	ENGAGE	ASSESS
ISR Observer	*ISR Observer*	*Swarm Coord.*	*Strategy Analyst*	*Tactical Planner*	*Effectors Agent*
Multi-sensor detection & track initiation	Target geolocation & classification	Multi-UAV persistent surveillance	ROE evaluation & nomination	COA generation & authorization	BDA observation & re-strike eval

OPERATOR CONTROL PLANE

GATE 1 — NOMINATION: Operator APPROVE / REJECT / RETASK verified targets

GATE 2 — COA AUTHORIZATION: Operator selects from 3 ranked Courses of Action (speed / P(Kill) / cost)

Three Autonomy Levels: MANUAL (operator decides all) | SUPERVISED (AI proposes, operator approves) | AUTONOMOUS (AI executes, operator monitors)

MULTI-SENSOR FUSION

Complementary fusion across all ISR modalities

Fused Confidence = 1 - PRODUCT(1 - ci) *with max-within-type deduplication*

EO/IR

Electro-Optical / Infrared
FMV capable
Day/night operations

SAR

Synthetic Aperture Radar
All-weather capability
Ground mapping

SIGINT

Signals Intelligence
RF spectrum analysis
Emitter geolocation

GMTI

Ground Moving Target Indicator
Velocity filtering

FMV

Full Motion Video
Persistent surveillance
Pattern-of-life analysis

- Unscented Kalman Filter (UKF) per-target with temporal decay
- Cross-sensor disagreement detection flags conflicting intelligence
- Probabilistic detection model (Pd) based on range, RCS, weather, sensor type
- Immutable dataclasses ensure data integrity across fusion pipeline

ORDER OF BATTLE

10 target types with unique behavior models

TARGET TYPE	BEHAVIOR	THREAT	PRIORITY
SAM	Stationary, emitting	30 km WEZ	CRITICAL
TEL	Shoot-and-scoot	Ballistic	CRITICAL
MANPADS	Ambush, flee on threat	5 km WEZ	HIGH
RADAR	Stationary, emitting	80-150 km det.	HIGH
C2 NODE	Stationary	Cmd & Control	HIGH
APC / IFV	Patrol	Direct fire	MEDIUM
ARTILLERY	Fire & reposition	Indirect fire	MEDIUM
CP (Command Post)	Stationary	HVT	MEDIUM
TRUCK / LOGISTICS	Patrol (slow)	Supply	LOW

VERIFICATION STATE MACHINE

DETECTED → CLASSIFIED → VERIFIED → NOMINATED (threat-adaptive thresholds, regression timeouts)

UAV OPERATIONS & SWARM COORDINATION

11 flight modes with autonomous swarm tasking

MODE	TGT REQ	DESCRIPTION
SEARCH	No	Circular loiter over assigned zone
FOLLOW	Yes	2 km orbit, visual contact
PAINT	Yes	1 km laser-designation orbit
INTERCEPT	Yes	1.5x speed, 300m danger-close
SUPPORT	Yes	Swarm sensor assist
VERIFY	Yes	Sensor contribution for verification
OVERWATCH	No	Elevated overwatch position
BDA	Yes	Post-strike damage assessment
REPOSITIONING	-	Auto-rebalance to coverage gap
RTB	-	Return to base (low fuel)

Swarm Engine

- Greedy UAV-to-target assignment with sensor-gap detection
- Priority: threat weight x verification gap x coverage need
- 120s task TTL with auto-release
- Idle-count guard for reserves
- Formation auto-adjusts per target type
- Zone imbalance drives repositioning
- Counter-UAS intercept mode

Counter-UAS: Tracks RECON, ATTACK, JAMMING, and EVADING enemy UAVs with dedicated intercept modes

OPERATOR INTERFACE

3D geospatial COP with real-time tactical overlays

CesiumJS 3D Globe
Real-time entity rendering
19 dedicated hooks
All entities on one COP

6 Map Modes
OPERATIONAL, COVERAGE,
THREAT, FUSION,
SWARM, TERRAIN

Multi-Sensor Camera
EO/IR, SAR, SIGINT, FUSION
SINGLE/PIP/SPLIT/QUAD
layouts

SIGINT Display
RF spectrum waterfall
Signal classification
Emitter geolocation

Strike Board
Nomination queue
COA comparison view
Two-gate HITL workflow

Battlespace Assessment
Threat clusters, coverage
gaps, zone heatmap,
movement corridors

THEATER CONFIGURATIONS

Pre-configured OPLAN scenarios, extensible via YAML

Romania Eastern Flank

NATO Article 5 Defense

BLUE FORCE

20x MQ-9 Reaper
24h endurance
EO/IR sensor suite

RED FORCE

3x SAM, 4x TEL, 8x TRUCK
2x CP, 6x MANPADS
2x RADAR, 1x C2 NODE
3x LOGISTICS

TERRAIN

Grassland / Clear

South China Sea

Maritime Domain Awareness

BLUE FORCE

15x P-8 Poseidon
10h endurance
Maritime radar @ 6km

RED FORCE

3x DESTROYER, 4x FRIGATE
6x MISSILE BOAT
2x SUBMARINE
12x FISHING VESSEL

TERRAIN

Maritime / Partly cloudy

Baltic States

NATO Forward Defense

BLUE FORCE

12x MQ-9 Reaper
20h endurance
EO/IR sensor suite

RED FORCE

5x SAM, 3x TEL
10x MANPADS, 4x RADAR
8x APC, 6x IFV, 4x MBT
2x C2 NODE, 5x LOGISTICS

TERRAIN

Urban-forest / Overcast

Theater configs are YAML-based — new OPLANs deployable in hours with custom Blue/Red ORBAT, ROE, weather, and terrain parameters.

RULES OF ENGAGEMENT & COMPLIANCE

Built for DoD Directive 3000.09 compliance

ROE ENGINE

- Theater-specific ROE loaded from YAML configs
- Per-target identity classification: HOSTILE, SUSPECT, NEUTRAL, FRIENDLY
- ROE action types: ENGAGE, OBSERVE_ONLY, WITHDRAW, REPORT
- Proportional response based on threat level and PID confidence
- All ROE evaluations logged with full audit trail
- Strategy Analyst agent enforces ROE at Gate 1

AUDIT & ACCOUNTABILITY

- Async JSONL event logging with daily rotation
- Every AI recommendation includes reasoning chain
- Two-gate HITL prevents unauthorized engagements
- Kill-chain tracking for After-Action Reviews (AARs)
- Structured logging via structlog across all subsystems
- Performance Auditor agent monitors decision quality

BATTLESPACE ASSESSMENT

Automated threat analysis and resource allocation

Threat Clustering

Auto-groups hostile assets into SAM_BATTERY, CONVOY, CP_COMPLEX, AD_NETWORK, and MIXED clusters for prioritized targeting

Coverage Gap Detection

Identifies under-surveilled zones and triggers autonomous UAV repositioning. Ensures persistent ISR across the AO.

Zone Threat Scoring

Per-zone threat level based on target types, count, and proximity. Drives ISR priority queue and asset allocation.

Movement Corridors

Predictive analysis identifies likely routes for shoot-and-scoot TELs and convoy movements. Enables preemptive positioning.

ISR Priority Queue

Urgency = threat weight x (1 - verification progress) x sensor gap. Optimizes limited ISR assets against verified threats.

Pattern Analysis

Historical activity analysis detects anomalies in convoy frequency, new facility construction, and operational tempo surges.

TECHNICAL SPECIFICATIONS

Open architecture built for interoperability

BACKEND

- Python 3.11+
- FastAPI + WebSocket
- 10Hz simulation loop
- Dual-client architecture
- Async JSONL logging
- Pydantic data contracts

FRONTEND

- React + TypeScript
- CesiumJS 3D Globe
- Blueprint UI (Dark Theme)
- Zustand State Mgmt
- Canvas drone cameras
- 75 UI components

AI / ML

- LangGraph / LangChain
- Multi-LLM fallback
- Unscented Kalman Filter
- Bayesian sensor fusion
- Pattern recognition
- Heuristic default mode

OPERATIONAL READINESS

Production-grade engineering metrics

475
Automated Tests
23 test files covering all subsystems

138
Python Modules
32,000+ lines of tested code

9
AI Agents
Complete F2T2EA coverage

30
WS Actions
Full bidirectional protocol

3
Theater Configs
Deployable scenarios

10Hz
Update Rate
Real-time tactical loop

DEPLOYMENT MODES

Three operational modes for any mission profile

INTERACTIVE C2

```
$ ./palantir.sh
```

Full operator-in-the-loop mode. All HITL gates active. Operator approves every nomination and COA. Recommended for live operations.

DEMO AUTO-PILOT

```
$ ./palantir.sh --demo
```

Autonomous F2T2EA execution with no human input. Auto-approves nominations (5s delay) and COAs (3s delay). Ideal for demonstrations and training.

HEADLESS BACKEND

```
$ ./palantir.sh --no-sim
```

Backend-only mode without drone video simulator. Lighter resource footprint for integration testing and CI/CD pipelines.

RECOMMENDED NEXT STEPS

- Schedule live demonstration with operational scenario of your choice
- Provide theater requirements for custom OPLAN configuration
- Coordinate sensor integration requirements for your ISR asset fleet

PALANTIR C2

AI-Enabled Command & Control

Accelerate the kill chain. Maintain the human edge.

REQUEST A LIVE DEMONSTRATION